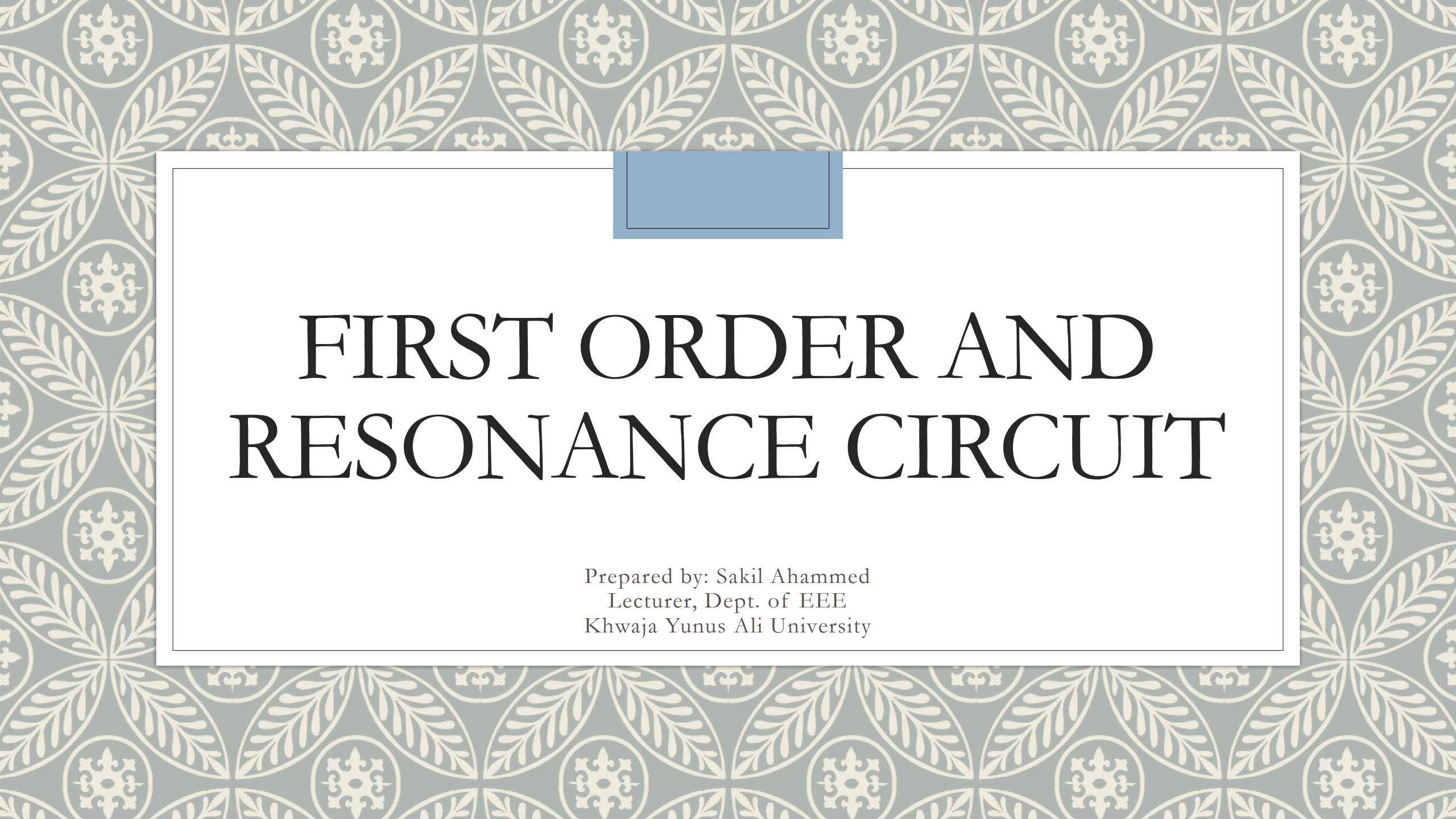


FIRST ORDER AND RESONANCE CIRCUIT

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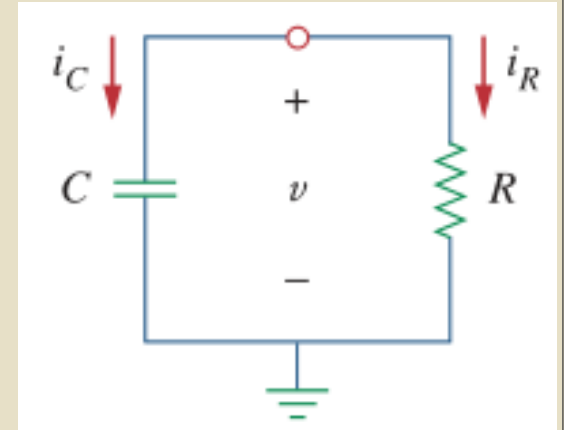
First Order Circuits- Introduction

- The only difference is that applying KCL to purely resistive circuits results in algebraic equations. While applying the laws to RC and RL circuits produces differential equations. The differential equations resulting from analyzing RC and RL circuits are of the first order. Hence, the circuits are known as first order circuits.
- A first order circuit is characterized by a first order differential equation.
- There are two ways available to excite those circuits.
- First: Initial condition of the storage elements in the circuits. Source free circuit, it is assumed that energy is initially stored in this capacitor or inductor. This energy causes current to flow in the circuit and gradually dissipated in the resistor to zero. It may have dependent sources.
- Second: Independent source connection.
- Typical applications of RL and RC circuit: Delay and relay circuits, a photoflash unit, and an automobile ignition circuit.

The source-Free RC circuit

- A source free RC circuit occurs when its dc source is suddenly disconnected. The energy stored in the capacitor is released to the resistors.
- Our objective is to determine the circuit response, here $v(t)$..(A circuit response is the manner in which the circuit reacts to an excitation).
- When the source was disconnected at $t=0$, capacitor was initially charge. Thus initial voltage is $v(0) = V_0$

And initial stored energy, $w(0) = \frac{1}{2} C V_0^2$
Applying KCL at the top node of the circuit,



Or,

Or,

Rearranging,

Integrating on both sides, we get

Thus,

And,

$$i_C + i_R = 0$$

$$C \frac{dv}{dt} + \frac{v}{R} = 0$$

$$\frac{dv}{dt} + \frac{v}{RC} = 0$$

$$\frac{dv}{v} = -\frac{1}{RC} dt$$

$$\ln v = -\frac{t}{RC} + \ln A \quad (\text{Here } A \text{ is the integration constant})$$

$$\ln \frac{v}{A} = -\frac{t}{RC}$$

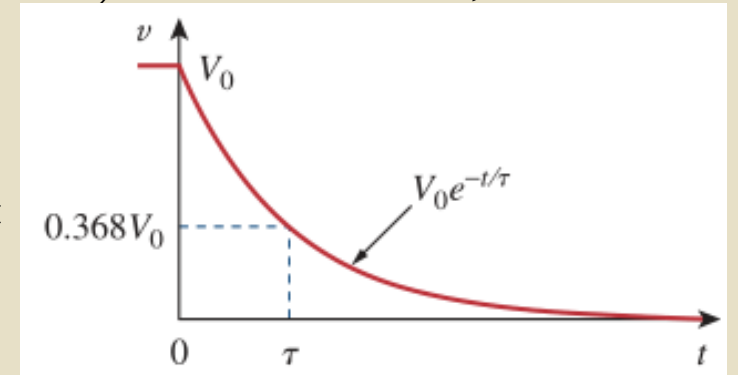
$$v(t) = A e^{-t/RC} = V_0 e^{-t/RC} \quad (\text{Initially } v(0) = A = V_0)$$

$$(\text{as, } i_C = C \frac{dv}{dt} \text{ \& } i_R = \frac{v}{R})$$

(This is the first order differential equation)

The source-Free RC circuit contd...

- The above equation shows that the voltage response of the RC circuit is an exponentially decay of the initial voltage.
- Since the response is due to the initial energy stored and the physical characteristics of the circuit and not due to some external voltage or current source, it is called the natural response of the circuit..
- “The natural response of a circuit refers to the behavior (in terms of voltage and current) of the circuit itself, with no external sources of excitation.”
- The graph shows that, initially stored voltage decreases exponentially towards zero with time.
- The rapidity with which the voltage decrease is expressed in terms of time constant denoted by τ (lowercase Greek letter tau).
- The time constant of a circuit is the time required for the response to decay to a factor of $1/e$ or 36.8 percent of its initial value.
- At $t = \tau$, $V_0 e^{-\tau/RC} = V_0 e^{-1} = 0.368 V_0$
- Thus, $\tau = RC$
- We can rewrite the equation as, $v(t) = V_0 e^{-t/\tau}$
- Note: $i_R(t) = \frac{v(t)}{R}$



The voltage response curve of RC circuit

Problems

In Fig. 7.5, let $v_C(0) = 15 \text{ V}$. Find v_C , v_x , and i_x for $t > 0$.

Example 7.1

Practice Problem 7.1

Refer to the circuit in Fig. 7.7. Let $v_C(0) = 60 \text{ V}$. Determine v_C , v_x , and i_o for $t \geq 0$.

Example 7.2

The switch in the circuit in Fig. 7.8 has been closed for a long time, and it is opened at $t = 0$. Find $v(t)$ for $t \geq 0$. Calculate the initial energy stored in the capacitor.

If the switch in Fig. 7.10 opens at $t = 0$, find $v(t)$ for $t \geq 0$ and $w_C(0)$.

Practice Problem 7.2

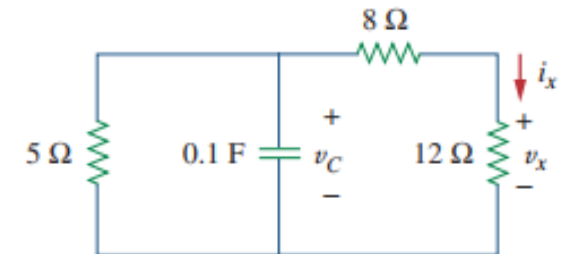


Figure 7.5

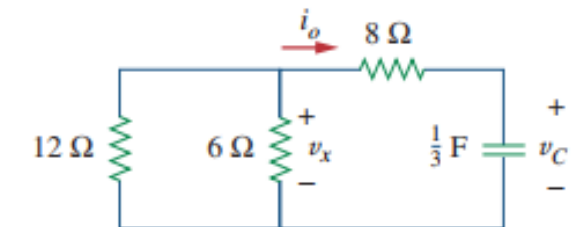


Figure 7.7

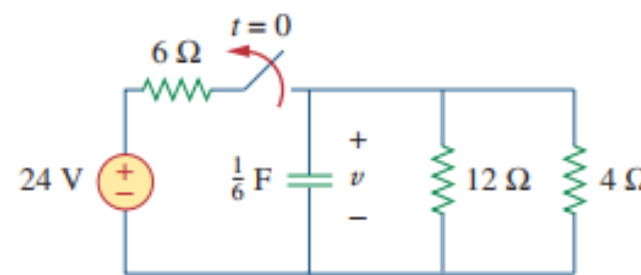


Figure 7.10

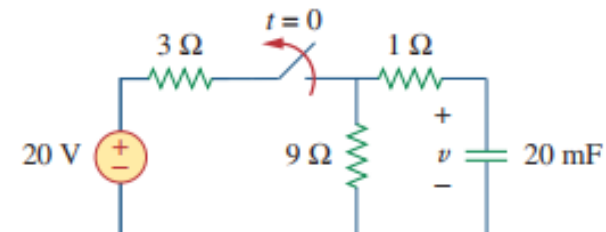


Figure 7.8

The source-free RL circuit

In this case, the circuit response is the current through the inductor as current cannot change Instantaneously in case of inductor.

At $t=0$, we assume the initial current through inductor is I_0 . or $i(0) = I_0$

With the corresponding energy stored in the inductor as, $w(0) = \frac{1}{2} L I_0^2$

Applying KVL around loop in the figure,

$$v_L + v_R = 0$$
$$L \frac{di}{dt} + Ri = 0$$

Or,

$$\frac{di}{dt} + \frac{Ri}{L} = 0$$

$$\int_{I_0}^{i(t)} \frac{di}{i} = - \int_0^t \frac{R}{L} dt$$

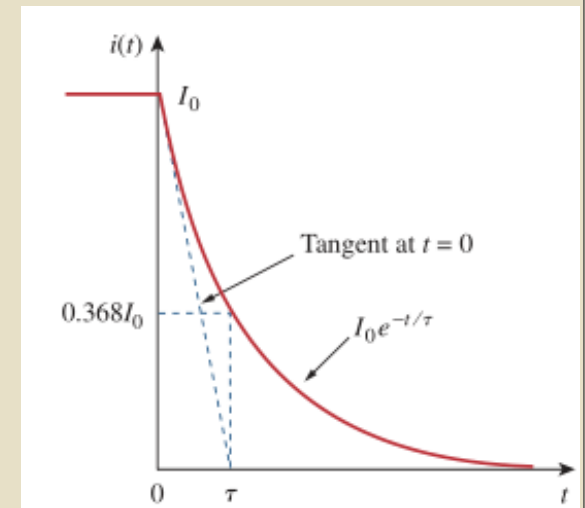
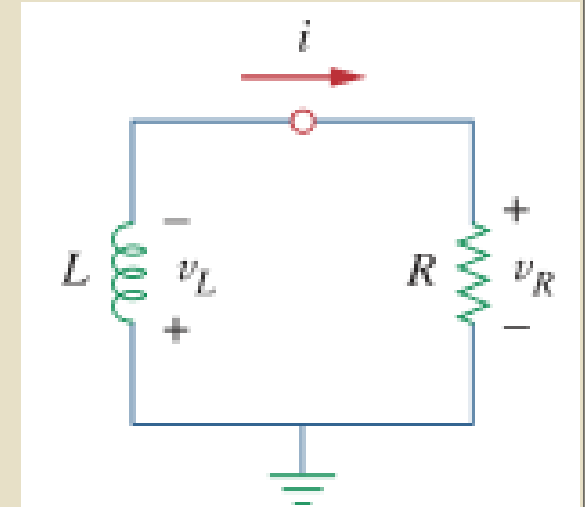
Or,

$$\ln \frac{i(t)}{I_0} = - \frac{Rt}{L}$$

Or,

$$i(t) = I_0 e^{-Rt/L}$$

It shows that the natural response of the RL circuit is an exponentially decay nature of the initial current.



RL circuit time constant

- Time constant of RL circuit is $\tau = \frac{L}{R}$ (Unit of τ is seconds)
- Thus, $i(t) = I_0 e^{-t/\tau}$
- $v_R(t) = iR$
- $p = v_R i$
- $w_R(t) = \int_0^t p(t) dt$

Problems

The switch in the circuit of Fig. 7.16 has been closed for a long time.

Example 7.4

At $t = 0$, the switch is opened. Calculate $i(t)$ for $t > 0$.

Practice Problem 7.4

For the circuit in Fig. 7.18, find $i(t)$ for $t > 0$.

Example 7.5

In the circuit shown in Fig. 7.19, find i_o , v_o , and i for all time, assuming that the switch was open for a long time.

Determine i , i_o , and v_o for all t in the circuit shown in Fig. 7.22.

Practice Problem 7.5

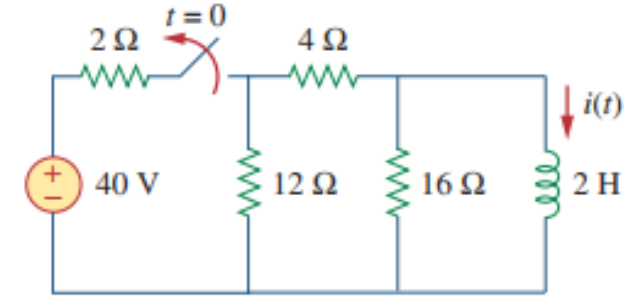


Figure 7.16

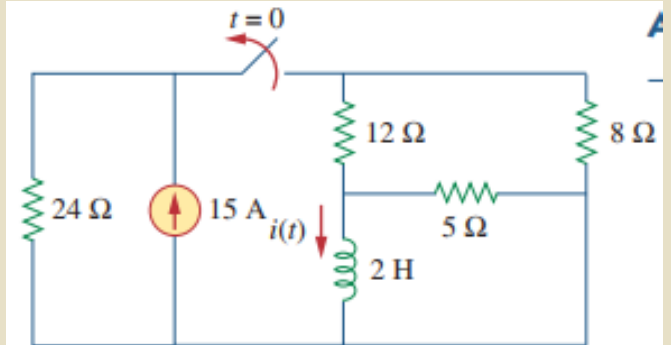


Figure 7.18

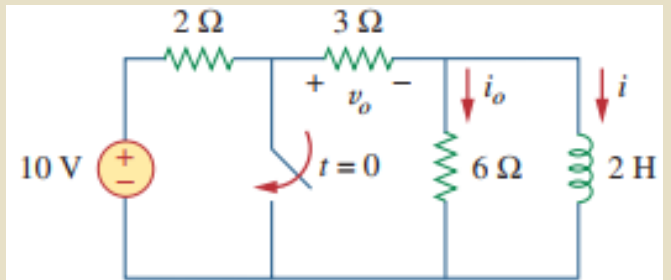


Figure 7.19

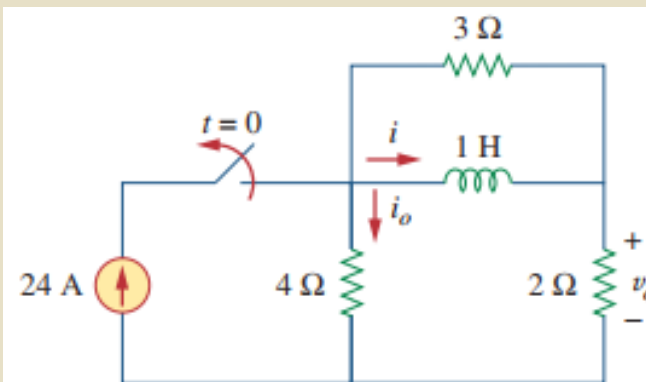
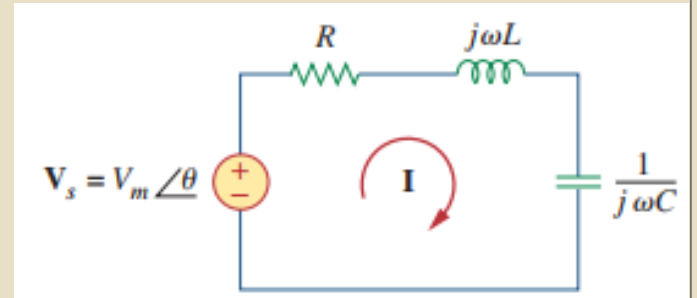


Figure 7.22

Resonance

Concept: The most prominent feature of the frequency response of a circuit may be the sharp peak (or the resonant peak) exhibited in the amplitude characteristic. Resonance occurs in any system that has a complex conjugate pair of poles; it is the cause of oscillations of stored energy from one form to another. Resonance occurs in any circuit that has at least one inductor and one capacitor.

- **Resonance is a condition in an RLC circuit in which the capacitive and inductive reactances are equal in magnitude, thereby resulting in a purely resistive impedance.**
- Resonance circuits are used in radio/tv receiver, filters and many more applications.
- Total Impedance of the network $Z = R + j(X_L - X_C)$
- In order to be a purely resistive circuit $(X_L - X_C)$ must be zero.
- Thus we get, $X_L = X_C$ (*Condition of resonance*)
- Suppose at frequency f_0 they become equal.
- We can write, $\omega_0 L = \frac{1}{\omega_0 C}$
- Or, $\omega_0 = \frac{1}{\sqrt{LC}}$ rad/s
- Finally, $f_0 = \frac{1}{2\pi\sqrt{LC}}$ Hz



Note that at resonance:

1. The impedance is purely resistive, thus, $Z = R$. In other words, the LC series combination acts like a short circuit, and the entire voltage is across R .
2. The voltage V_s and the current I are in phase, so that the power factor is unity.
3. The magnitude of the transfer function $H(\omega) = Z(\omega)$ is minimum.
4. The inductor voltage and capacitor voltage can be much more than the source voltage.

Resonance contd...

The average power dissipated by the RLC circuit is $p(\omega) = \frac{1}{2} I^2 R$

The highest power dissipated occurs at resonance, when $I = V_m/R$

$$\text{So, } p(\omega_0) = \frac{1}{2} \frac{V_m^2}{R}$$

At certain frequencies $\omega = \omega_1, \omega_2$, The dissipated power is half of the maximum value

$$\text{That is, } P(\omega_1) = P(\omega_2) = \frac{(\frac{V_m}{\sqrt{2}})^2}{2R} = \frac{V_m^2}{4R}$$

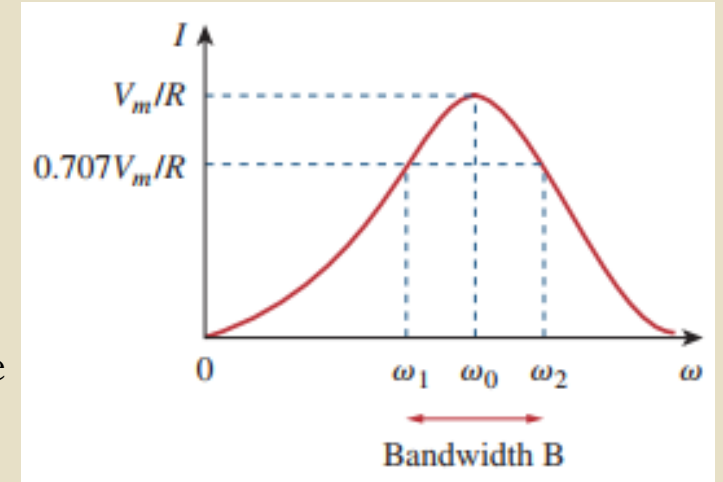
Here, ω_1 and ω_2 are called the half power frequencies.

The half power frequencies can be obtained by setting Z equal to $\sqrt{2}R$.

$$\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2} = \sqrt{2}R$$

$$\omega_1 = -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

$$\omega_2 = \frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$



Resonant frequency, $\omega_0 = \sqrt{\omega_1 \omega_2}$

The height of the curve depends on R , but the width of the curve depends on bandwidth (B), which is the difference between the two half-power frequencies,

$$B = \omega_2 - \omega_1$$

The sharpness of the resonance in a resonant circuit is measured quantitatively by the quality factor Q . At resonance the reactive energy in the circuit oscillates between the inductor and the capacitor.

Resonance contd...

The quality factor relates the maximum or peak energy stored to the energy dissipated in the circuit per cycle of oscillation.

$$Q = 2\pi \frac{\text{Peak energy stored in the circuit}}{\text{Energy dissipated by the circuit in one period at resonance.}}$$

$$Q = 2\pi \frac{\frac{1}{2}LI^2}{\frac{1}{2}I^2R(1/f_0)} = \frac{2\pi f_0L}{R}$$

Or,

$$Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 CR}$$

$$\omega_1 \cong \omega_0 - \frac{B}{2}, \quad \omega_2 \cong \omega_0 + \frac{B}{2}$$

Quality factor is dimensionless.

$$B = \frac{R}{L} = \frac{\omega_0}{Q}$$

$$\text{or } B = \omega_0^2 CR$$

The quality factor of a resonant circuit is the ratio of its resonant frequency to its bandwidth.

If $Q \geq 10$, it is high quality factor.

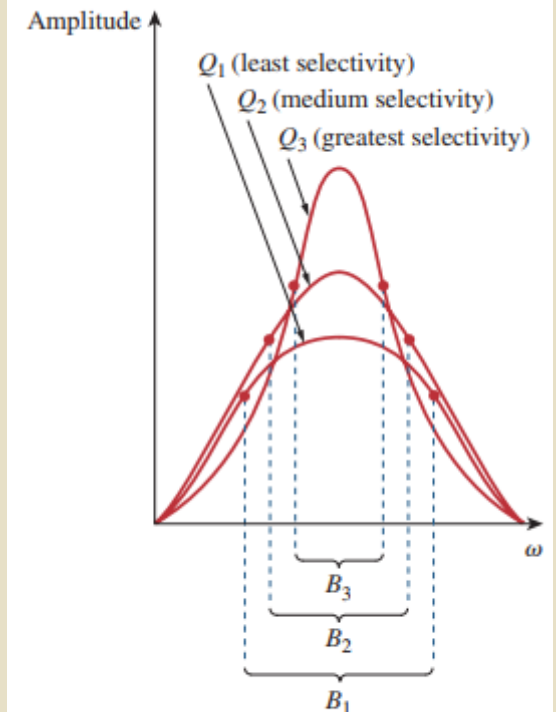


Figure 14.23

The higher the circuit Q , the smaller the bandwidth.

Characteristic

Series circuit

Resonant frequency, ω_0

$$\frac{1}{\sqrt{LC}}$$

Quality factor, Q

$$\frac{\omega_0 L}{R} \text{ or } \frac{1}{\omega_0 RC}$$

Bandwidth, B

$$\frac{\omega_0}{Q}$$

Half-power frequencies, ω_1, ω_2

$$\omega_0 \sqrt{1 + \left(\frac{1}{2Q}\right)^2} \pm \frac{\omega_0}{2Q}$$

For $Q \geq 10$, ω_1, ω_2

$$\omega_0 \pm \frac{B}{2}$$

Problems

In the circuit of Fig. 14.24, $R = 2\ \Omega$, $L = 1\text{ mH}$, and $C = 0.4\ \mu\text{F}$.

(a) Find the resonant frequency and the half-power frequencies. (b) Calculate the quality factor and bandwidth. (c) Determine the amplitude of the current at ω_0 , ω_1 , and ω_2 .

Example 14.7

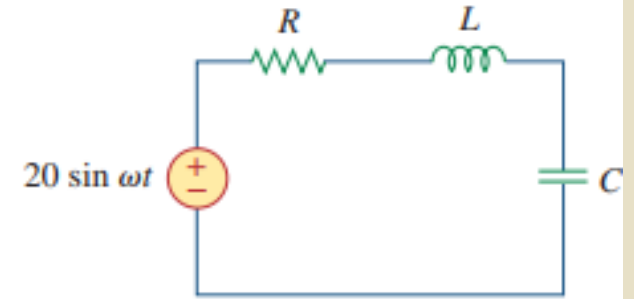


Figure 14.24

Practice Problem 14.7

A series-connected circuit has $R = 4\ \Omega$ and $L = 25\text{ mH}$. (a) Calculate the value of C that will produce a quality factor of 50. (b) Find ω_1 , ω_2 , and B . (c) Determine the average power dissipated at $\omega = \omega_0$, ω_1 , ω_2 . Take $V_m = 100\text{ V}$.

Answer: (a) $0.625\ \mu\text{F}$, (b) 7920 rad/s , 8080 rad/s , 160 rad/s , (c) 1.25 kW , 0.625 kW , 0.625 kW .

Anybody can become angry-that is easy; but to be angry with the right person, and to the right degree, and at the right time, and for the right purpose, and in the right way-that is not within everybody's power and is not easy.

Aristotle